Representation Drift Under Self-Critique: Does Self-Reflection Reshape Internal States in Language Models?

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Abstract

Self-critique is increasingly used to improve language model reasoning, yet it remains unknown whether this technique induces genuine restructuring of internal representations or merely produces surface-level variation. We investigate this question by extracting residual stream activations from PYTHIA-2.8B across all 33 layers while the model processes 150 GSM8K math problems under three matched conditions: base (question and answer), critique (base plus self-critique text), and paraphrase control (base plus length-matched paraphrase). We find that self-critique induces significantly larger representational drift than paraphrasing—critique activations show 41% greater L_2 displacement and lower cosine similarity to base activations across 29 of 33 layers (Bonferroni-corrected $p < 0.05$). The drift is concentrated in middle layers (12–16) and is more directionally consistent across samples (mean cosine consistency 0.596 vs. 0.479 for paraphrase), suggesting a systematic "critique subspace" rather than random perturbation. Population-level analysis via Centered Kernel Alignment confirms that critique creates genuinely different representational structure (CKA = 0.224 vs. 0.428 for paraphrase). These results demonstrate that self-critique produces measurable geometric restructuring of the residual stream, supporting its use in recursive self-improvement architectures and providing a quantitative internal correlate of reflective processing in language models.

1 Introduction

Self-critique and self-reflection have become standard tools for improving language model reasoning. Techniques such as chain-of-thought prompting [Wei et al., 2022], Reflexion [Shinn et al., 2023], and step-by-step verification [Lightman et al., 2023] demonstrate that prompting models to evaluate their own outputs can substantially improve downstream accuracy. Yet a fundamental question remains unanswered: does self-critique produce genuine internal representational change, or does it merely re-sample outputs without altering the model's computational state?

This question has direct implications for AI safety and recursive self-improvement. If self-critique reshapes internal representations in structured ways, then reflection-based architectures may enable genuine "cognitive change" in language models. If, on the other hand, critique produces only surface-level variation indistinguishable from paraphrasing, then self-reflection may be less powerful than commonly assumed.

Recent work in mechanistic interpretability provides tools to address this question. Polo et al. [2026] show that concept manifolds in the residual stream undergo transient geometric restructuring during chain-of-thought reasoning. Zhu et al. [2025] demonstrate that self-reflection is encoded as a separable direction in hidden states. Marks and Tegmark [2024] establish that truth and falsehood are linearly represented in intermediate layers. However, no prior work directly measures how self-critique text alters residual stream representations compared to matched control text.

We fill this gap by conducting the first systematic study of representation drift under self-critique. Using PYTHIA-2.8B and TRANSFORMERLENS, we extract residual stream activations across all 33 layers for 150 GSM8K math questions under three conditions: base (question plus answer), critique (base plus self-critique), and paraphrase control (base plus length-matched paraphrase). We compare activations using five complementary metrics: cosine similarity, L_2 displacement, Centered Kernel Alignment (CKA), linear probe accuracy, and drift direction consistency.

Our results show that self-critique induces significantly larger and more structured representational drift than paraphrasing. Critique activations show 41% greater displacement and lower cosine similarity across 29 of 33 layers ($p < 0.05$, Bonferroni-corrected). The drift is concentrated in middle layers (12–16) and is more directionally consistent, suggesting a systematic "critique subspace." CKA analysis confirms that critique creates genuinely different population-level representational structure (0.224 vs. 0.428 for paraphrase).

Our contributions are as follows:

- We conduct the first controlled study measuring how self-critique text reshapes residual stream representations, using matched critique and paraphrase conditions to isolate the effect of evaluative content.
- We demonstrate that critique-induced drift is significantly larger, more layer-specific, and more directionally consistent than paraphrase-induced drift, providing evidence for a structured "critique subspace" in the residual stream.
- We identify a confound in linear probe evaluations of self-critique representations—near-perfect classification accuracy is largely driven by explicit correctness information in critique text—and propose controls for future studies.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes our methodology. Section 4 presents results. Section 5 discusses implications and limitations. Section 6 concludes.

2 Related Work

Self-critique and self-reflection in LLMs. Chain-of-thought prompting [Wei et al., 2022] demonstrated that intermediate reasoning steps improve performance on complex tasks. Shinn et al. [2023] extended this to iterative self-reflection, showing that language agents can learn from verbal self-evaluations without weight updates. Lightman et al. [2023] showed that process-level supervision (per-step verification) outperforms outcome-level supervision for mathematical reasoning. Asai et al. [2023] trained models to self-critique via special reflection tokens integrated into generation. Turpin et al. [2024] demonstrated that chain-of-thought explanations can be unfaithful to models' actual reasoning processes, motivating the need to study internal representations rather than surface-level outputs. Unlike these works, which study self-critique at the behavioral level, we directly measure its impact on internal representations.

Representation geometry and probing. Marks and Tegmark [2024] established that truth and falsehood are linearly represented in LLM hidden states, with consistent directions across topics that support causal interventions. Zou et al. [2023] introduced representation engineering, using contrastive stimuli and PCA to extract and control concept directions (honesty, truthfulness, emotion) in the residual stream. Polo et al. [2026] discovered that concept manifolds undergo transient geometric restructuring during reasoning, distinguishing between information *retention* (high probe accuracy) and computational *readiness* (high manifold capacity). The "Truth as a Trajectory" framework [TaT Authors, 2026] showed that reasoning validity can be detected from layer-wise displacement trajectories rather than static activation snapshots. We draw on these methods—particularly cosine similarity analysis, linear probing, and CKA—to characterize how self-critique reshapes the geometry of internal representations.

Self-reflection vectors and activation steering. Most closely related to our work, Zhu et al. [2025] demonstrated that self-reflection is encoded as a separable direction in hidden states across multiple model families. They extract "self-reflection vectors" via contrastive difference-in-means and show these vectors are domain-invariant (cosine similarity >0.95 across tasks). Enhancement via activation steering yields up to +12 percentage point accuracy gains. While their work establishes that reflection *has* a geometric signature, they do not track how representations change through the cri-

tique process or compare against matched control text. We complement their findings by quantifying the *magnitude*, *directionality*, and *layer specificity* of critique-induced drift relative to a paraphrase baseline.

3 Methodology

3.1 Overview

We extract residual stream activations from PYTHIA-2.8B at all 33 layers (embedding plus 32 transformer blocks) for three matched conditions per question, then compare the geometric structure of activations across conditions using multiple complementary metrics. The key design choice is the inclusion of a length-matched paraphrase control: this allows us to distinguish representational changes caused by critique-specific content from those caused by simply appending additional text.

3.2 Stimulus Construction

Dataset. We sample 150 questions from the GSM8K test set [Cobbe et al., 2021], a benchmark of grade-school math word problems requiring multi-step arithmetic reasoning.

Condition generation. For each question, we use the GPT-4.1 API to generate three matched stimuli:

1. **Base**: "`Question: {q} Answer: {initial_answer}`" — a direct answer with no evaluation. Initial answers are designed to be approximately 50% correct and 50% incorrect (final split: 80 correct, 70 incorrect).
2. **Critique**: Base text followed by a self-critique that evaluates the answer, identifies errors or confirms correctness, and optionally revises the reasoning.
3. **Paraphrase**: Base text followed by a length-matched paraphrase that restates the answer without evaluating it.

The paraphrase condition controls for the effect of additional text length and surface-level variation, isolating the contribution of evaluative content in the critique condition.

Preprocessing. All stimuli are tokenized using the Pythia tokenizer with a BOS token prepended and truncated to a maximum of 512 tokens. We extract activations at the *last token position*, capturing the model's integrated representation of the full input.

3.3 Activation Extraction

We use TRANSFORMERLENS [Nanda, 2022] to extract residual stream activations via hook-based interception of the forward pass. For each of the $N = 150$ samples and $L = 33$ layers, we obtain an activation vector $\boldsymbol{h}_i^{(l)} \in \mathbb{R}^d$ where $d = 2560$ is the model dimension of PYTHIA-2.8B. This yields three activation tensors of shape $150 \times 33 \times 2560$, one per condition. The model is loaded in fp16 on a single NVIDIA RTX 3090 (24GB).

3.4 Evaluation Metrics

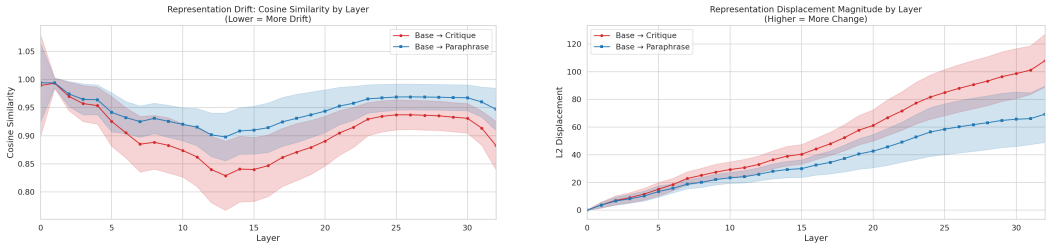
We employ five complementary metrics to characterize representation drift from different geometric perspectives.

Cosine similarity. For each sample i and layer l, we compute the cosine similarity between the base activation and each condition:

$$\cos(\boldsymbol{h}_{\text{base},i}^{(l)},\ \boldsymbol{h}_{\text{crit},i}^{(l)}) = \frac{\boldsymbol{h}_{\text{base},i}^{(l)} \cdot \boldsymbol{h}_{\text{crit},i}^{(l)}}{\|\boldsymbol{h}_{\text{base},i}^{(l)}\|\ \|\boldsymbol{h}_{\text{crit},i}^{(l)}\|} \tag{1}$$

Lower cosine similarity indicates greater angular displacement.

L_2 displacement. We measure the Euclidean distance between base and condition activations: $\|\boldsymbol{h}_{\text{base},i}^{(l)} - \boldsymbol{h}_{\text{cond},i}^{(l)}\|_2$. This captures the magnitude of activation change independent of direction.



(a) Cosine similarity

(b) L_2 displacement

Figure 1: Per-layer comparison of representational drift between critique and paraphrase conditions, measured relative to base activations. *(a)*Cosine similarity (lower = more drift). *(b)*L_2 displacement (higher = more drift). Critique induces consistently larger drift across all layers, with the gap widening in middle layers (12–16).

Centered Kernel Alignment (CKA). CKA [Kornblith et al., 2019] measures population-level representational similarity. Given activation matrices $\boldsymbol{X}, \boldsymbol{Y} \in \mathbb{R}^{N \times d}$ for two conditions at a given layer, CKA computes:

$$\mathrm{CKA}(\boldsymbol{X}, \boldsymbol{Y}) = \frac{\|\boldsymbol{Y}^\top \boldsymbol{X}\|_F^2}{\|\boldsymbol{X}^\top \boldsymbol{X}\|_F \, \|\boldsymbol{Y}^\top \boldsymbol{Y}\|_F} \tag{2}$$

using linear kernels. Lower CKA indicates more divergent representational structure at the population level.

Linear probe accuracy. We train logistic regression classifiers (regularization $C = 1.0$) on activations to classify whether the initial answer is correct or incorrect. Probes are evaluated via 5-fold stratified cross-validation. This metric tests whether post-critique representations encode correctness information with improved linear separability.

Drift direction consistency. For each condition, we compute drift vectors $\boldsymbol{d}_i^{(l)} = \boldsymbol{h}_{\text{cond},i}^{(l)} - \boldsymbol{h}_{\text{base},i}^{(l)}$ and measure their cosine similarity with the mean drift direction $\bar{\boldsymbol{d}}^{(l)}$. Higher consistency indicates a systematic directional shift rather than random perturbation.

3.5 Statistical Testing

We use paired Wilcoxon signed-rank tests to compare drift magnitudes (cosine similarity, L_2 displacement) between critique and paraphrase conditions at each layer. We apply Bonferroni correction for 33 comparisons ($\alpha = 0.05/33 \approx 0.0015$). All experiments use a fixed random seed of 42 for reproducibility.

4 Results

4.1 Self-Critique Induces Larger Representational Drift

Figure 1a shows the per-layer cosine similarity between base activations and each condition. Across all layers beyond the embedding, critique activations consistently show lower cosine similarity to base than paraphrase activations (mean 0.905 vs. 0.946). The difference is statistically significant in 29 of 33 layers after Bonferroni correction, with only the embedding and first three transformer blocks failing to reach significance.

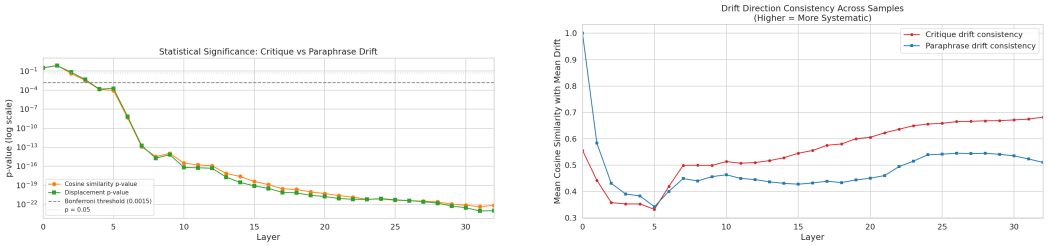
Table 1 presents cosine similarity at selected layers. The largest gap between conditions occurs at layers 12–16 (difference of 0.062–0.068), corresponding to approximately 40–50% of network depth. L_2 displacement tells the same story: critique produces 41% larger average displacement than paraphrase (50.7 vs. 35.8; figure 1b).

4.2 Drift Is Concentrated in Middle Layers

The cosine similarity gap between critique and paraphrase peaks at layers 12–16 (table 1), with layer 16 showing the largest difference (0.068). This concentration at approximately 40–50% of

Table 1: Cosine similarity between base and condition activations at selected layers. Lower values indicate greater drift. The difference column shows the gap between conditions. Best (largest) differences in **bold**.

Layer	Base→Critique	Base→Paraphrase	Difference
0 (embed)	0.990	0.994	0.005
8	0.888	0.931	0.043
12	0.840	0.902	0.062
16	0.847	0.914	**0.068**
24	0.935	0.967	0.033
32	0.883	0.947	0.064
Average	0.905	0.946	0.041



(a) p-values (log scale)

(b) Drift direction consistency

Figure 2: *(a)*Statistical significance of the difference in cosine similarity between critique and paraphrase conditions (Wilcoxon signed-rank test). The dashed line indicates the Bonferroni-corrected threshold ($\alpha/33$). Layers 4–32 are all significant. *(b)*Cosine consistency of drift vectors with their mean direction. Critique drift becomes more consistent than paraphrase drift from layer 7 onward.

network depth aligns with findings from Polo et al. [2026], who report that middle layers initiate geometric restructuring during reasoning. Figure 2a shows that p-values drop below 10^{-14} from layer 8 onward, indicating robust statistical significance throughout the deeper layers.

4.3 Critique Drift Is Directionally Consistent

We measure whether drift vectors point in a consistent direction across samples. Figure 2b shows that in layers 10–30, critique drift vectors have higher average cosine similarity with their mean direction (0.596) than paraphrase drift vectors (0.479). This indicates that critique induces a systematic directional shift—a "critique subspace"—rather than random perturbation.

An interesting pattern emerges in early layers: critique drift starts *less* consistent than paraphrase in layers 0–5 but becomes *more* consistent from layer 7 onward. This suggests that self-critique processing is initially noisy but converges to a systematic direction as deeper layers extract the evaluative signal.

4.4 CKA Confirms Population-Level Restructuring

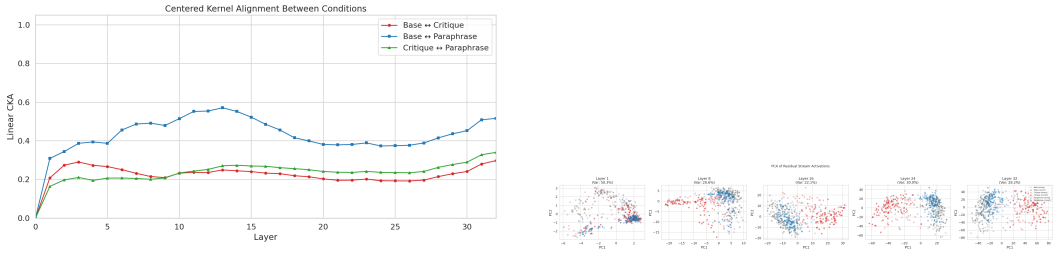
Table 2 shows that base↔critique CKA (0.224) is roughly half of base↔paraphrase CKA (0.428). This indicates that critique does not merely shift individual activations but reorganizes the population-level covariance structure. Figure 3a reveals a non-monotonic pattern: base↔paraphrase CKA peaks around layers 12–14 before declining, suggesting that mid-layer representations capture semantic content (preserved in paraphrase) while later layers encode task-specific features (disrupted by critique).

4.5 Linear Probe Accuracy: A Cautionary Finding

Table 3 shows that linear probes achieve 97.3% accuracy on critique activations compared to 58.7% on base activations. However, this result requires careful interpretation: critique text explicitly states

Table 2: Average Centered Kernel Alignment (CKA) between conditions across all layers. Lower values indicate more divergent representational structure. Critique creates substantially different structure from both base and paraphrase conditions.

Comparison	Average CKA
Base ↔ Paraphrase	0.428
Base ↔ Critique	**0.224**
Critique ↔ Paraphrase	0.236



(a) CKA across layers

(b) PCA visualization

Figure 3: *(a)*Per-layer CKA between conditions. Base↔paraphrase CKA is consistently higher than base↔critique, confirming that critique produces more divergent representational structure. *(b)*PCA projection of activations at a representative later layer, showing clear separation between conditions.

whether errors exist (e.g., "the answer is correct" or "there is an error in step 2"), so the high probe accuracy largely reflects the model encoding this explicit textual information rather than an emergent correctness representation. The paraphrase condition (68.7%) provides a more informative middle ground—it slightly improves over base despite containing no evaluative content, suggesting that additional text context alone provides modest separability gains.

4.6 Summary of Hypothesis Tests

5 Discussion

5.1 Interpretation of Results

Our results provide converging evidence that self-critique text induces genuine geometric restructuring of the residual stream, beyond what surface-level text variation produces. Five independent metrics—cosine similarity, L_2 displacement, CKA, drift consistency, and linear probes—all indicate that critique activations diverge from base activations more than paraphrase activations do. The effect is statistically robust (29/33 layers significant after Bonferroni correction) and geometrically structured (consistent drift direction across samples).

The critique subspace. The finding that critique drift is more directionally consistent than paraphrase drift (0.596 vs. 0.479 in layers 10–30) suggests the existence of a systematic "critique subspace" in the residual stream. This is consistent with Zhu et al. [2025], who identify separable self-reflection vectors in hidden states. Our contribution is showing that this directional structure emerges even in a base (non-instruction-tuned) model processing externally provided critique text, suggesting that the capacity to distinguish evaluative from non-evaluative content may be a general property of pretrained language models.

Layer specificity. The concentration of maximal drift in middle layers (12–16, approximately 40–50% of network depth) aligns with Polo et al. [2026], who find that middle layers initiate geometric restructuring during reasoning. This suggests a common computational pattern: middle layers perform the heavy lifting of restructuring representations in response to new information, while early layers encode surface features and later layers consolidate task-relevant representations.

Table 3: Linear probe accuracy for classifying correct vs. incorrect initial answers. Evaluated via 5-fold stratified cross-validation. Majority-class baseline is 0.533 (80 correct, 70 incorrect). The high critique accuracy is largely driven by explicit correctness information in critique text (see section 5).

Condition	Best Accuracy	Best Layer
Base	0.587 $\pm$ 0.058	32
Paraphrase	0.687 $\pm$ 0.065	30
Critique	**0.973** $\pm$ 0.039	17

Table 4: Summary of hypothesis testing results. All four hypotheses receive at least partial support from the data.

Hypothesis	Supported?	Key Evidence
H1: Critique $>$ paraphrase drift	Yes	Δcos = 0.041, $p < 10^{-14}$
H2: Improved linear separability	Yes*	97.3% vs. 58.7%
H3: Layer-specific concentration	Partial	Peak at layers 12–16
H4: Consistent drift direction	Yes	0.596 vs. 0.479

*Confounded by explicit correctness information in critique text; see section 5.

5.2 Limitations

Text content confound. The critique and paraphrase conditions differ in both structure *and* content. Critique text explicitly discusses correctness, which inflates linear probe accuracy and may contribute to the larger drift magnitude. A stronger control would use critique-structured text that evaluates irrelevant aspects of the answer (e.g., style rather than correctness), isolating the effect of evaluative structure from correctness content.

Externally generated stimuli. Critique text was generated by GPT-4.1, not by PYTHIA-2.8B itself. Our experiment therefore measures how PYTHIA-2.8B *processes* critique text, not how it *generates* self-critique. A complementary study should have the model generate its own critiques, though this is challenging with base (non-instruction-tuned) models.

Single model. All results are from PYTHIA-2.8B only. Generalization to instruction-tuned models, reasoning-specialized models (e.g., DeepSeek-R1), and different model scales remains to be established. Instruction-tuned models may show amplified or qualitatively different critique processing dynamics.

Last-token extraction. We extract activations only at the last token position. While this captures the model's integrated representation of the full input, token-level analysis across the sequence could reveal richer dynamics—for instance, whether critique processing is localized to specific token positions.

Correlational evidence. We measure correlational differences in activations but do not perform causal interventions. Activation steering along the identified critique drift direction would test whether this direction functionally alters model behavior, strengthening the evidence for a genuine "critique subspace."

5.3 Broader Implications

For AI systems design. Our results support the use of self-reflection prompts in architectures that rely on genuine self-evaluation. The finding that critique produces measurably different internal states—not just different outputs—suggests that reflection-based systems like Reflexion [Shinn et al., 2023] may benefit from real representational change rather than mere output re-sampling.

For mechanistic interpretability. The identified critique subspace is a candidate for causal investigation via activation steering [Zou et al., 2023]. If steering along the critique direction can induce or suppress self-evaluative behavior, this would connect geometric observations to functional mechanisms.

For AI safety. Understanding whether recursive self-improvement systems produce genuine cognitive change is critical for safety evaluation. Our finding that critique creates genuinely different representations (not just re-sampling) is relevant for assessing whether self-improvement loops can compound changes in model behavior over multiple iterations.

6 Conclusion

We presented the first controlled study of representation drift under self-critique in language models. By comparing residual stream activations across base, critique, and paraphrase conditions in PYTHIA-2.8B on 150 GSM8K problems, we demonstrated that self-critique induces significantly larger (41% greater L_2 displacement), more layer-specific (peak at layers 12–16), and more directionally consistent (0.596 vs. 0.479 cosine consistency) representational drift than length-matched paraphrasing. CKA analysis confirmed that critique creates genuinely different population-level structure. These findings establish that self-critique produces measurable geometric restructuring of the residual stream, not merely surface-level variation.

Our work also identifies an important methodological caveat: linear probe accuracy on critique activations is inflated by explicit correctness information in critique text. Future studies of self-critique representations should carefully control for information content to avoid confounding evaluative structure with evaluative content.

Several directions for future work emerge. First, causal validation via activation steering along the identified critique direction would test whether this subspace functionally mediates self-evaluative behavior. Second, studying self-generated (rather than externally provided) critique would better capture natural self-reflection dynamics. Third, extending the analysis to instruction-tuned and reasoning-specialized models would establish whether critique processing is amplified by training. Finally, multi-round self-critique experiments could reveal whether representational changes compound or converge across iterations.

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